

 BANNARI AMMAN INSTITUTE OF TECHNOLOGY Stay Ahead	BANNARI AMMAN INSTITUTE OF TECHNOLOGY (An Autonomous Institution Affiliated to Anna University, Chennai) SATHYAMANGALAM - 638 401
	Approved Questions S2 Semester
22CH203 – ENGINEERING CHEMISTRY II	

Q. No.	Questions
	Unit 1
1.	A laboratory constructs an electrochemical setup using zinc and copper metals placed in their respective ionic solutions. The zinc electrode gradually dissolves while copper ions get deposited as solid copper. A voltmeter connected between the electrodes shows a steady potential difference under standard conditions. The system operates without any external power supply. The experimenter observes directional electron flow through the external circuit. Analyze the system to determine the origin of the observed potential difference and infer how chemical reactions establish a driving force for electron movement across the circuit while maintaining charge balance within the system.
2.	A researcher compares two electrochemical cells, one operating under standard conditions and another with altered ion concentrations. The measured potential differs significantly between the two systems. Both systems use the same electrode materials but vary in solution composition. The researcher attempts to relate this observation to thermodynamic principles governing electrochemical reactions. Determine how variations in chemical conditions influence electrode potential and justify the thermodynamic basis for potential generation in electrochemical systems.
3.	An electrochemical system is represented using symbolic notation showing two half-cells separated by a salt bridge. One half-cell contains a metal immersed in its ions, while the other contains a different metal-ion pair. The notation includes arrows indicating reaction direction. A student misinterprets the sign of the cell potential and predicts incorrect electron flow direction. Interpret the correct representation of the electrochemical cell and infer how sign conventions determine the direction of electron flow and feasibility of the reaction.
4.	A student assembles a half-cell using a metal electrode dipped in an electrolyte but is unable to measure its absolute potential directly. The student then connects it to a standard hydrogen electrode to obtain a measurable value. The setup produces a stable voltage reading. Analyze why a reference electrode is necessary and determine how it enables measurement of electrode potential.
5.	In an experiment, a metal electrode is immersed in its ionic solution, and a potential develops at the interface. The system shows no external current flow but still maintains a measurable potential difference across the interface. Interpret the origin of this interfacial potential and infer the role of equilibrium in its formation.

6.	A closed electrochemical circuit shows continuous electron flow through a wire while ions move through the electrolyte. The system maintains neutrality despite ongoing reactions. Determine how ionic movement supports sustained electron flow in the circuit.
7.	Two different metals are connected without an electrolyte, and no current is observed despite a difference in their properties. Infer why an electrolyte is essential for current generation in electrochemical systems.
8.	A laboratory electrochemical cell is constructed using two electrodes immersed in electrolyte solutions of different ionic strengths, and the system is allowed to reach equilibrium under open-circuit conditions while maintaining constant temperature and pressure, after which the potential difference is measured and later compared with the voltage recorded when a small current is drawn through an external circuit, revealing a noticeable difference between the two readings. Analyze the observed difference between the measured potentials under open-circuit and load conditions, and determine the underlying electrochemical principles governing this variation while justifying the role of internal resistance and system equilibrium.
9.	An industrial electrochemical setup operates under varying environmental conditions where temperature increases gradually and reactant concentration changes due to continuous consumption, while pressure variations also occur in a gas electrode system, leading to fluctuations in measured cell potential over time. Determine how variations in concentration, temperature, and pressure influence the observed EMF and justify the combined effect of these parameters using electrochemical reasoning.
10.	A researcher evaluates two electrochemical cells, one designed to operate reversibly under controlled slow reactions and another subjected to rapid discharge with significant energy losses, and compares their efficiencies while attempting to relate electrical output to thermodynamic feasibility. Analyze the differences between the two cells in terms of reversibility and infer how these differences affect the relationship between EMF and Gibbs free energy.
11.	A standard hydrogen electrode is coupled with an unknown metal electrode under standard conditions, and the measured EMF is used to determine the relative position of the metal in an electrochemical series for predicting reaction feasibility. Interpret how the measured EMF can be used to position the metal in the electrochemical series and infer its reactivity.
12.	An electrochemical cell is analyzed under standard conditions where all reactants are maintained at unit activity, and the EMF value is recorded to compare it with values obtained under non-standard laboratory conditions. Determine the significance of standard EMF and justify its role in comparing electrochemical systems.
13.	A student measures EMF using a voltmeter directly across a working electrochemical cell that is simultaneously supplying current to a load, leading to inaccurate readings. Identify the limitation in the measurement approach and infer a more appropriate method for accurate EMF determination.

14.	An electrochemical reaction shows a positive EMF value during operation, and the researcher attempts to infer the spontaneity and energy change associated with the reaction. Interpret the significance of positive EMF in relation to Gibbs free energy and reaction feasibility.
15.	A metal electrode is immersed in an aqueous electrolyte containing both cations and anions, where the metal surface acquires a net charge due to electron transfer. The nearby ions rearrange themselves forming a structured region near the interface. The compactness and distribution of ions vary depending on thermal motion and electrostatic attraction. Experimental observations indicate that the simple rigid-layer assumption fails to explain the diffuse nature of ion distribution at higher temperatures. Analyze how different theoretical models interpret the structure of the electrical double layer at the electrode–electrolyte interface and justify which model best reconciles both compact and diffuse charge distributions under varying conditions.
16.	An electrochemical cell shows variation in interfacial potential when the concentration of electrolyte changes. Measurements reveal that the potential drop is not linear but varies significantly close to the electrode surface. The electric field strength is highest near the electrode and gradually decreases into the bulk solution. Determine how the potential gradient develops across the electrical double layer and evaluate its relationship with charge distribution and capacitance at the interface.
17.	In a system involving polar solvents, it is observed that solvent molecules orient themselves near the electrode surface even before significant ion accumulation occurs. Additionally, certain ions are found to specifically adsorb onto the electrode, altering the effective surface charge. Interpret the combined influence of solvent orientation and specific ion adsorption on the effective structure and behavior of the electrical double layer.
18.	During an electrochemical reaction, the rate of electron transfer is observed to change when the electrolyte concentration is varied. The structure of the double layer shifts accordingly, influencing the accessibility of reacting species to the electrode surface. Analyze how changes in the electrical double layer influence electrode kinetics and reaction rates.
19.	An interface between a metal and electrolyte exhibits measurable changes in surface tension when an external potential is applied. These changes correlate with variations in surface charge density. Evaluate the relationship between surface charge density and electrocapillary effects at the interface.
20.	A system shows that increasing ionic strength reduces the thickness of the diffuse layer significantly. Infer how ionic strength affects the structure of the electrical double layer
21.	An electrode exhibits higher capacitance when placed in a concentrated electrolyte compared to a dilute one. Determine the reason for increased double layer capacitance in concentrated solutions

22.	A lithium-ion battery under high discharge is observed to show uneven ion distribution near the electrodes, where lithium ions accumulate on one side while depletion occurs on the other, despite the presence of an electric field driving ions across the electrolyte. The electrolyte viscosity is moderate, and no external stirring is applied, while temperature remains constant throughout the process. Analyze how the combined effects of migration and diffusion govern the observed ion distribution and determine the dominant transport mechanism under the given conditions while justifying your reasoning using electrochemical transport principles.
23.	An electrochemical cell using an aqueous electrolyte exhibits reduced efficiency when electrolyte concentration is increased beyond a certain level, even though higher ion availability is expected to enhance conductivity. Measurements reveal increased ion interactions and reduced mobility at higher concentrations. Evaluate the relationship between electrolyte concentration and conductivity and determine why increased concentration leads to reduced efficiency beyond an optimal point
24.	A fuel cell shows voltage drop during prolonged operation, where analysis indicates buildup of reaction products near the electrode surface, causing a local change in ion concentration and affecting charge transport. Determine how concentration polarization develops in this system and assess its impact on overall cell performance using transport principles.
25.	An electrolyte solution under an applied electric field shows that cations move faster than anions due to smaller size and higher mobility, contributing unevenly to current flow. Analyze how transport numbers arise in this system and determine their significance in current distribution.
26.	A battery exhibits increased internal resistance as it ages, even though electrode materials remain unchanged, and electrolyte conductivity slightly decreases over time. Evaluate the factors contributing to increased internal resistance and infer its effect on battery performance.
27.	An electrochemical system experiences enhanced ion transport when stirred externally, even though no changes are made to electric field or concentration. Determine the role of convection in this system and justify its effect on transport enhancement.
28.	In an electrolyte under steady conditions, both electric field and concentration gradient are present, influencing ion flux simultaneously. Interpret how the Nernst-Planck framework integrates these effects to govern ion movement.
29.	A laboratory team constructs an electrochemical setup using a zinc rod immersed in zinc sulfate solution and a copper rod immersed in copper sulfate solution, connected via a salt bridge containing potassium nitrate. During operation, they observe electron flow through the external circuit and ion migration through the salt bridge, but also note a gradual decrease in voltage over time due to changes in ion concentration near the junction. Analyze the working mechanism of this electrochemical cell by determining the role of each component and infer the reasons for voltage variation with respect to junction behavior and ion movement within the system.

30.	An industrial system uses an electrolytic cell to deposit a metal coating onto a conductive surface. The setup includes an inert platinum electrode as the anode and a metal ion solution. Operators notice uneven deposition and increased energy consumption during prolonged operation. Evaluate the operational inefficiencies in this electrolytic cell by correlating electrode type, reaction pathways, and energy requirements, and recommend how system design influences performance outcomes.
31.	A technician records the notation of an electrochemical cell as $\text{Fe} \mid \text{Fe}^{2+} \parallel \text{Ag}^+ \mid \text{Ag}$ and attempts to interpret the direction of electron flow and identify half-cell reactions. However, confusion arises regarding the representation and sequence of components in the notation. Interpret the given cell notation by determining electrode roles, direction of electron transfer, and reconstruct the half-cell reactions while justifying the conventions used in representation.
32.	A student studies the function of a salt bridge and replaces it with a simple metallic wire between two half-cells. The system initially shows current but quickly stops functioning. Determine the cause of system failure by analyzing the role of ionic conduction and infer why the replacement disrupts sustained operation.
33.	A fuel cell operates using hydrogen and oxygen gases with porous electrodes and an electrolyte medium. Over time, efficiency drops due to electrode surface contamination. Analyze the operational principle of the fuel cell and infer how electrode conditions influence overall efficiency.
34.	A lead-acid battery is used in an automobile and undergoes repeated charging and discharging cycles, leading to gradual performance degradation. Infer the chemical changes responsible for this degradation and relate them to electrode reactions.
35.	An electrochemical setup uses a gas electrode where hydrogen gas is bubbled over a platinum surface in contact with an acidic solution. Determine the role of platinum and infer how this electrode enables redox reactions.
36.	A galvanic cell is constructed using a zinc electrode immersed in 1 M Zn^{2+} solution and a copper electrode immersed in 1 M Cu^{2+} solution. The standard electrode potentials are given as $\text{Zn}^{2+}/\text{Zn} = -0.76 \text{ V}$ and $\text{Cu}^{2+}/\text{Cu} = +0.34 \text{ V}$. The system operates under standard conditions, and a student attempts to determine whether the reaction is spontaneous and how the electron flow occurs in the circuit. Analyze the given electrochemical system and determine the direction of electron flow, overall cell reaction, and spontaneity by integrating the electrode potentials and electrochemical principles.
37.	A concentration cell is built using the same metal electrode but with ion concentrations of 0.01 M and 1 M respectively. The system operates at 298 K, and the student uses the Nernst equation to estimate the cell potential, considering logarithmic dependence on ion concentration. Determine how the difference in concentration influences the EMF and evaluate the role of the reaction quotient in predicting the direction of the cell operation.

38.	An electrochemical cell operates at 298 K where the EMF is measured as 0.59 V for a reaction involving transfer of 2 electrons. The student is required to determine the equilibrium constant for the reaction using thermodynamic relationships linking EMF and equilibrium. Formulate the relationship between EMF and equilibrium constant and determine the magnitude of the equilibrium constant using the provided data.
39.	A student observes that increasing temperature slightly alters the EMF of an electrochemical cell, even though concentrations remain constant. The reaction involves heat absorption during operation. Analyze how temperature variation influences cell potential and infer its effect based on thermodynamic reasoning.
40.	Two metals A and B, are compared using the electrochemical series, where A is placed higher than B. The student is asked to predict displacement reactions in aqueous solutions. Determine how the electrochemical series can be used to infer reaction feasibility and metal displacement behavior.
41.	A metal structure exposed to moisture shows gradual deterioration, and electrochemical data suggests oxidation tendencies of the material. Evaluate how electrode potentials help in predicting corrosion susceptibility in the given system with an example.
42.	A sensor device uses an electrochemical cell to detect ion concentration changes in a solution, and its output voltage varies accordingly. Determine how cell potential variation enables sensing functionality in such devices.
	Unit 2
1.	An electrochemical device operates in a cold region where low temperature and diluted electrolyte reduce performance. The observed EMF is lower than standard values, affecting energy output. Engineers aim to improve efficiency under these conditions. Analyze how concentration variations influence the electrical energy output of the cell and determine strategies to improve performance using thermodynamic relationships.
2.	A laboratory cell shows declining voltage after repeated use due to solvent evaporation and slight heating. These changes shift the system away from standard conditions. The output energy becomes inconsistent over time. Diagnose the causes of reduced EMF and infer how these deviations affect Gibbs free energy and overall efficiency.
3.	A galvanic cell with known EMF and electron transfer number is analyzed under reversible conditions. The researcher wants to estimate maximum electrical work. The system follows thermodynamic principles. Determine the relationship between EMF and Gibbs free energy and infer how it indicates spontaneity and energy output.

4.	An electrochemical system operates at constant temperature but varying electrolyte concentrations. The EMF changes with concentration differences. The energy output becomes variable. Analyze how concentration changes affect EMF and infer the impact on energy output.
5.	Different electrode materials are tested under standard conditions. Each combination produces a different EMF. The goal is to identify the most efficient pair. Determine how standard electrode potentials influence EMF and energy efficiency.
6.	A cell shows reduced output at higher temperature compared to optimal conditions. The system deviates slightly from equilibrium. The energy produced decreases. Infer how temperature variation affects Gibbs free energy and EMF.
7.	A system uses measured EMF to estimate electrical energy output. The cell operates under known conditions. The output depends on voltage. Determine how EMF relates to electrical energy output of the cell.
8.	A laboratory demonstrates an electrolytic cell where external electrical energy drives chemical reactions. This setup differs from galvanic cells where reactions occur spontaneously. Students observe how energy input influences reaction direction and electron flow. Infer the procedural flow of energy and reaction direction in an electrolytic cell compared to a galvanic cell
9.	A rural lighting project relies on primary cells, leading to frequent replacements and increased waste generation. The lack of rechargeability raises sustainability concerns. The project team considers switching to rechargeable alternatives to improve efficiency. Assess the suitability of replacing primary cells with secondary cells by considering sustainability and operational efficiency.
10.	A research team compares two electrochemical cells where one has high energy density but low power density, and the other has low energy density but high power density. The application requires rapid bursts of energy rather than long-term supply. Efficiency under load conditions is also considered in performance evaluation. Determine which cell is more suitable for high-power applications by analytically interpreting energy density and power density relationships.
11.	A portable electronics company designs compact devices requiring stable voltage output using electrochemical cells. Engineers analyze how electrodes, electrolytes, and separators interact to maintain consistent performance. Variations in material properties affect ion transfer and internal resistance. The system must ensure durability while minimizing energy loss. Integrate the roles of key cell components to infer how their interaction determines energy output consistency in portable devices.

12.	An electric vehicle manufacturer compares lithium-ion, lead-acid, and dry cells to select a battery system for large-scale production. Lithium-ion offers high energy density but higher cost, while lead-acid provides lower cost but reduced efficiency and lifespan. Dry cells lack rechargeability, limiting scalability for vehicles. Engineers evaluate trade-offs involving power density, lifecycle, efficiency, and thermal stability under operational demands. Prioritize the most scalable cell technology by analyzing trade-offs between energy density, cost, efficiency, and lifecycle performance, and justify the selection logically.
13.	A manufacturing plant observes inconsistent voltage output and rapid efficiency loss in electrochemical cells used for industrial sensors. The cells contain metal electrodes, liquid electrolytes, and separators, but performance varies across batches. Engineers suspect issues such as electrode degradation, electrolyte contamination, or separator malfunction affecting ion flow. Environmental exposure and assembly inconsistencies may also influence redox reactions and energy output. Analyze the possible causes of performance degradation by correlating component-level failures with electrochemical principles and infer the most critical fault affecting energy output.
14.	A remote healthcare unit in a mountainous region must select an electrochemical energy source to power diagnostic equipment continuously without frequent maintenance access. The team evaluates primary and secondary cells considering rechargeability, reaction reversibility, and environmental stability. Constraints include limited transportation, fluctuating temperatures, and the need for consistent energy output over long durations. Engineers analyze energy density, efficiency, and electrode durability to ensure reliable and sustainable operation. Determine the most suitable electrochemical cell classification for the healthcare unit by integrating energy storage requirements, reaction reversibility, and operational constraints, and justify the decision through systematic reasoning.
15.	A reversible electrochemical cell is used as a reference system in thermodynamic studies. Interpret why reversible cells are preferred in thermodynamic analysis and justify their role.
16.	A cell operates under standard conditions, and its EMF is calculated using standard electrode potentials. Determine the relevance of standard EMF in assessing reversibility and infer its thermodynamic significance.
17.	An electrochemical cell shows deviation from predicted potential values when operated in real-world conditions with high current flow and internal resistance. Analyze the limitations affecting reversibility in practical electrochemical cells and justify their impact on performance.

18.	A scientist studies electrode reactions in a cell where no current is drawn, and both oxidation and reduction occur at equal rates. The system appears stable with no net chemical change. Infer the state of the system and determine how electrode reactions contribute to reversibility under these conditions.
19.	An electrochemical system is designed to extract maximum electrical energy from a spontaneous redox reaction. The process is compared under two conditions: one where reactions occur rapidly with energy losses, and another where reactions proceed slowly maintaining equilibrium. Evaluate which condition ensures maximum work output and reconcile this with the concept of reversibility in electrochemical cells, supporting your conclusion with thermodynamic reasoning.
20.	A Daniell cell is constructed using Zn and Cu electrodes with their respective sulfate solutions. The concentration of Zn^{2+} ions is gradually increased while Cu^{2+} concentration remains constant. The experiment is conducted under controlled conditions, and the potential difference is continuously monitored. Determine how the change in Zn^{2+} concentration influences the cell potential using appropriate electrochemical principles, and justify the direction of variation by integrating equilibrium concepts.
21.	A research laboratory is testing an electrochemical cell where both electrodes are maintained under identical temperature and pressure while allowing ion exchange through a salt bridge. The system is allowed to operate very slowly so that equilibrium is nearly maintained at all times. The engineers observe no heat loss and maximum electrical work output during operation. However, they are unsure whether the cell fully satisfies thermodynamic reversibility conditions. Analyze the given system and determine whether the electrochemical cell qualifies as reversible by evaluating the thermodynamic conditions and operational characteristics. Justify your reasoning by integrating equilibrium behavior and energy considerations.
22.	A comparison between two cells shows one maintaining constant voltage under ideal conditions while another shows fluctuating voltage and energy loss during operation under real-world conditions. Analyze how operational behavior differentiates reversible and irreversible cells in terms of energy efficiency.

23.	A small electronic device battery shows slight heating and reduced output voltage when used for extended durations, indicating minor but noticeable inefficiencies in energy delivery. Determine how such small inefficiencies indicate irreversible behavior and infer their effect on device performance.
24.	An industrial battery system operates continuously and shows a sequence of losses starting from activation barriers to concentration imbalances and finally resistive heating under load conditions. Formulate the logical sequence of processes that convert an ideal electrochemical cell into an irreversible one during operation.
25.	A laboratory cell operating under ideal conditions is compared with a real system where entropy production is observed due to energy dissipation and heat generation during operation, especially under sustained current flow. Integrate entropy production with energy loss mechanisms to infer why practical efficiency deviates from theoretical predictions.
26.	A grid-scale storage system compares theoretical energy output with actual measured output under varying loads and finds increasing discrepancy at higher currents due to resistive heating and kinetic limitations within the electrochemical cell. Evaluate how increasing current leads to deviation from thermodynamic efficiency and recommend whether such a system can be considered reversible under practical conditions.
27.	An electric vehicle battery experiences gradual efficiency loss over repeated charging cycles where side reactions form unwanted deposits on electrodes and measurable overpotential increases with time, reducing the effective voltage output under load conditions. Analyze how side reactions and overpotential lead to irreversible behavior and infer their role in reducing both efficiency and lifespan of the battery system.
28.	A portable power unit used in remote field operations shows decreasing voltage output during continuous usage and operators notice that the system heats up while delivering current, and further testing indicates that internal resistance and polarization effects increase as load demand rises under non-ideal environmental conditions. Determine how internal resistance and polarization jointly contribute to classifying the system as irreversible and justify their impact on practical efficiency during continuous operation.
29.	A technician must maintain a fleet of batteries used in telecommunications infrastructure where reliability is critical. The batteries experience moderate usage but must remain ready for sudden discharge events. Environmental conditions are controlled, but operational discipline varies. Assess the professional practices required to maintain reversibility and ensure consistent performance in such systems.

30.	A charging system applies constant current initially and later switches to constant voltage mode in lithium-ion batteries to ensure safe charging. This staged approach is designed to optimize ion movement and prevent damage. The system transitions automatically based on voltage thresholds. Determine the logical sequence of this charging architecture and infer its role in maintaining reversibility.
31.	A battery system delivers 90 Wh during discharge but requires 110 Wh input during charging due to energy losses. The system operates under repeated cycles, and engineers aim to quantify efficiency and identify implications for long-term operation under repeated cycling conditions. Determine the charge-discharge efficiency and interpret its implications for energy losses and system performance.
32.	A hybrid energy storage system combines lithium-ion batteries with supercapacitors to manage peak loads in an industrial setup. During operation, batteries handle sustained energy delivery while supercapacitors manage sudden spikes. The system is designed to reduce stress on battery cycling and improve overall efficiency. Observations indicate improved cycle life but slightly increased system complexity. Integrate the role of hybridization in enhancing charge-discharge efficiency and infer how it improves reversibility within the battery system.
33.	An electric vehicle manufacturer compares two battery management strategies: one prioritizes full discharge to maximize range, while the other restricts discharge to 60% capacity to enhance battery lifespan. Field data shows that vehicles using full discharge deliver higher immediate performance but require battery replacement sooner, whereas restricted discharge vehicles show longer cycle life but reduced usable energy per cycle. The company must determine the optimal trade-off between performance and longevity. Analyze the trade-offs involved in both strategies and recommend an optimal approach by evaluating efficiency, cycle life, and electrochemical reversibility considerations.
34.	A lead-acid battery system used in backup power for a hospital begins to show reduced discharge capacity and incomplete charging even after extended charging periods. Electrolyte concentration measurements indicate irregular variations across cells, and frequent deep discharge cycles have been recorded during emergency usage. Additionally, the battery operates in a high-temperature environment due to poor ventilation. Engineers are tasked with diagnosing the root cause of reduced reversibility and efficiency degradation. Evaluate the operational conditions and infer the primary electrochemical causes of performance degradation, justifying how each factor contributes to loss of reversibility and reduced cycle efficiency.

35.	A renewable energy storage system in a remote microgrid relies on a bank of lithium-ion batteries that undergo frequent charge and discharge cycles due to intermittent solar input and fluctuating load demand across the day. The system operator observes that although the batteries initially delivered expected energy output, over time the usable capacity has declined and charging requires longer durations under similar conditions. Temperature variations between day and night are significant, and the depth of discharge varies depending on energy demand peaks. The operator must redesign the charging strategy to maintain reversibility and prolong system efficiency under these fluctuating environmental and operational conditions. Analyze the given system behavior and determine an optimized charging and discharging strategy that preserves electrode reversibility while minimizing efficiency losses, integrating relevant electrochemical and operational factors.
36.	A rural electrification project requires long-duration energy storage but faces budget constraints and maintenance limitations. Identify a suitable storage approach and justify its relevance to the scenario.
37.	A data center experiences frequent overheating in battery backup systems during peak load operations, raising concerns about system reliability. Determine the cause and recommend a conceptual solution to mitigate the issue.
38.	A manufacturing firm aims to reduce environmental impact by replacing conventional battery systems with eco-friendly alternatives. Current systems rely on rare metals and generate disposal challenges. Analyze how material selection influences sustainability and propose key considerations for eco-friendly design.
39.	A startup is designing energy storage for portable medical devices where compact size and quick charge-discharge cycles are critical. Traditional batteries fail to meet rapid energy delivery requirements efficiently. Infer which emerging technology is most suitable and justify its selection based on operational requirements.

40.	A research lab is comparing solid-state batteries with conventional lithium-ion batteries for use in aerospace applications. The focus is on safety, energy density, and operational reliability under extreme temperature variations and mechanical stress conditions. Formulate a comparative assessment and infer how solid-state technology addresses the limitations of traditional systems in this context.
41.	An electric vehicle manufacturer observes that battery packs degrade rapidly under high-temperature operating conditions, leading to reduced vehicle range and safety concerns. Investigations reveal electrode material instability and electrolyte decomposition during repeated charging cycles, especially in urban driving environments with frequent acceleration and braking. Determine the underlying material-related challenges contributing to degradation and justify how they impact both performance and safety of the system.
42.	A national electric grid operator is planning large-scale integration of renewable energy sources such as solar and wind, but faces inconsistencies in supply due to intermittency and storage inefficiencies. Existing lithium-ion battery systems are deployed but show limitations in energy density and cost, leading to high infrastructure expenses and reduced feasibility for long-term storage. Engineers are evaluating alternative storage solutions while considering scalability and lifecycle performance under fluctuating load demands. Analyze the constraints imposed by current electrochemical storage systems in the given scenario and determine how these limitations influence large-scale renewable integration decisions.
Unit 3	
1.	A student observes that when zinc reacts in a solution, it loses electrons while another species gains electrons. Determine the distinction between oxidation and reduction processes in this context and relate it to corrosion.
2.	During a laboratory experiment, a metal strip is partially immersed in an electrolyte solution, and corrosion is observed only in the immersed region. Infer the essential components required for corrosion to occur in this setup and justify their interaction.

3.	A metallic bridge exposed to coastal conditions shows uniform rusting across large areas. Engineers classify this as a common type of corrosion influenced by environmental exposure. Analyze the type of corrosion occurring and determine the contributing factors that lead to its uniform nature.
4.	A buried pipeline experiences corrosion at certain sections where soil moisture and composition vary. Investigations reveal that some regions act differently despite being part of the same metal structure. Determine how variations in environmental conditions lead to formation of corrosion cells within the same metal and infer the roles of different regions in the process.
5.	In an industrial cooling system, two dissimilar metals are connected and immersed in circulating water containing dissolved salts. Over time, one metal deteriorates significantly while the other remains relatively unaffected. Engineers aim to understand the role of electrode potentials in this degradation. Evaluate how differences in electrode potentials influence corrosion behavior in the system and determine which metal is more likely to corrode while justifying the electron transfer process.
6.	An aluminum component used in an aircraft structure remains stable under normal atmospheric conditions but begins to degrade when exposed to a saline environment. Engineers observe that despite aluminum's reactivity, it often resists corrosion unless specific conditions disrupt its surface. Interpret the thermodynamic basis that governs the stability and corrosion tendency of aluminum and determine why environmental changes alter its behavior despite its inherent reactivity.
7.	A steel storage tank in a chemical plant is exposed to moist air containing dissolved oxygen and trace acidic vapors over a long period. The outer surface begins to show reddish-brown deposits, and localized thinning is observed. The plant engineer notes that certain regions corrode faster due to uneven exposure and suspects electrochemical activity at the surface. Analyze the underlying electrochemical processes governing the corrosion observed in the tank and determine how different regions of the metal surface function within a corrosion cell while justifying the role of environmental conditions.
8.	A metal structure immersed in water with high dissolved oxygen shows faster corrosion compared to the same structure in deaerated water. Infer the role of dissolved oxygen in influencing electrochemical corrosion.

9.	During corrosion testing, a metal exhibits reduced corrosion rate after applying an external current that opposes the natural electrochemical reactions occurring on its surface. Analyze how polarization influences corrosion rate in this context.
10.	A metal surface exposed to stagnant water develops regions with limited oxygen and regions with high oxygen concentration. Over time, pits begin to form in the low oxygen areas. Determine how differential oxygen availability leads to localized corrosion and infer the type of corrosion cell formed.
11.	A metallic component operating in an acidic solution experiences increased corrosion rate when temperature rises. Simultaneously, the system shows reduced resistance to current flow due to increased ionic mobility. Interpret the combined influence of temperature and electrolyte conductivity on the corrosion rate of the system.
12.	Two metals, zinc and copper, are placed in an electrolyte solution and electrically connected. Standard electrode potentials indicate zinc has a more negative potential compared to copper. After some time, zinc shows significant material loss while copper remains largely unaffected. Evaluate the role of standard electrode potentials in predicting corrosion tendency and justify the observed behavior of the metals in this system.
13.	An industrial storage tank made of iron is exposed to water containing dissolved oxygen and salts. Over time, reddish-brown deposits form on the surface. Chemical analysis identifies Fe^{2+} ions and hydroxide ions as intermediates in the corrosion process, and the overall reaction appears to involve both oxidation and reduction steps. Integrate the half-cell reactions involved in the corrosion process and formulate the overall electrochemical reaction responsible for rust formation in this system.
14.	A steel pipeline buried underground shows uneven corrosion patterns, with certain regions dissolving faster than others despite being made of the same material. Soil moisture, oxygen availability, and ionic content vary along the pipeline, leading to localized electrochemical activity. Investigations reveal formation of anodic and cathodic regions due to environmental heterogeneity, and measurable potential differences exist across segments of the metal surface. Analyze the electrochemical basis for the formation of anodic and cathodic regions in the given system and determine how these localized regions drive corrosion progression across the pipeline.

15.	An electrochemical cell shows ion accumulation near one electrode when voltage is applied. The concentration near the electrode becomes significantly higher than the bulk. Interpret the cause of ion accumulation and its relation to electric potential.
16.	A thin metallic wire carries steady current and operates under stable temperature. Over time, minor resistance increase is detected due to atomic displacement. Infer the underlying mechanism responsible for resistance change and relate it to ion movement.
17.	A cooling system uses fluid flow over a metallic surface while an electric potential is applied across it. Ion movement is influenced not only by electric field but also by fluid motion. Unexpected corrosion patterns emerge. Determine how convection interacts with migration and diffusion to alter corrosion behavior.
18.	A metallic coating on a structural component is exposed to an electrolyte under an applied voltage. Ions begin moving through the electrolyte and across the interface. Over time, thinning of the coating is observed at specific regions. Analyze the role of ionic drift and diffusion in causing selective material loss in the coating.
19.	In a semiconductor device, copper interconnects are subjected to alternating current conditions. Although current direction changes, gradual degradation is still observed. Material transport is influenced by both concentration gradients and transient electric fields. Device performance declines due to structural instability at microscopic scales. Formulate how multiple mass transport mechanisms collectively contribute to degradation under alternating current conditions and justify their relative significance.
20.	A buried pipeline carrying conductive fluid is exposed to stray electrical currents from nearby power systems. Over time, localized corrosion spots appear where current enters and exits the metal surface. Ionic species in the surrounding soil show directional movement aligned with potential gradients. The corrosion rate varies across different regions of the pipeline. Determine how ionic transport mechanisms interact with electric potential gradients to influence spatial variation in corrosion intensity along the pipeline.

21.	A microelectronic interconnect made of aluminum operates under high current density for prolonged durations. Over time, voids are observed near the cathode end while hillocks form near the anode. The system is maintained at elevated temperature, increasing atomic mobility. Engineers suspect that directional mass transport is occurring due to electric field influence combined with thermal activation. The failure leads to circuit discontinuity. Analyze the governing transport mechanism responsible for the observed material redistribution and determine how electric field and temperature jointly influence this degradation process in the interconnect.
22.	Two identical metal rods are placed in different solutions: one neutral and one strongly acidic. The acidic solution shows rapid corrosion with gas evolution. Infer the role of electrolyte composition in modifying electron transfer reactions and corrosion intensity.
23.	A metal sample is heated in an oxygen-rich furnace, and a uniform oxide layer forms across its surface. The growth rate slows down over time. Engineers consider diffusion and electron transfer limitations. Determine how electron transfer contributes to the decreasing rate of oxidation over time.
24.	A protective coating on a metal surface develops microscopic defects, allowing small regions to be exposed to a corrosive electrolyte. These exposed areas begin to corrode faster than the surrounding protected surface. Engineers analyze the system to understand how electron flow is concentrated in these regions. Analyze how localized defects influence electron transfer distribution and infer their role in accelerating corrosion kinetics.
25.	A corrosion engineer studies two metals exposed separately to humid air and dry air at room temperature. The metal in humid air corrodes rapidly, while the one in dry air shows negligible corrosion. The engineer suspects the presence of an electrolyte film is critical for electron transfer continuity. Interpret how the presence of moisture alters electron transfer pathways and justify its impact on corrosion rate differences.

26.	In an acidic industrial cleaning system, metal equipment is exposed to a low pH solution without significant dissolved oxygen. Corrosion is still observed, and hydrogen gas evolution is detected at certain regions of the metal surface. Engineers recognize that corrosion is occurring even in the absence of oxygen and suspect an alternative cathodic reaction mechanism. Analyze the hydrogen evolution mechanism as a cathodic process and determine how electron transfer sustains corrosion in oxygen-deficient environments.
27.	A marine structure made of iron is partially submerged in seawater. Different regions of the metal surface exhibit varying oxygen concentrations due to uneven exposure to air and water flow. Some regions corrode faster, forming rust, while others remain relatively intact. Engineers identify that electrochemical reactions are occurring due to the presence of an electrolyte, and localized electron flow is driving the process. Determine how electron transfer between anodic and cathodic regions drives wet corrosion and infer the role of oxygen concentration differences in accelerating localized corrosion.
28.	An industrial steel pipeline carrying hot gases operates continuously at elevated temperatures in a chemical plant. Over time, engineers observe the formation of oxide scales on the metal surface even though the environment is completely dry. The oxide layer thickness increases with temperature, and the rate of corrosion appears to change depending on the diffusivity of ions through the oxide layer. The plant team suspects that electron transfer still plays a critical role even in the absence of moisture. They need to understand how oxidation progresses under these conditions and what governs the rate of material degradation. Analyze the electron transfer mechanism governing corrosion under dry conditions and determine how ionic and electronic movements collectively influence the oxidation rate of the metal surface.
29.	A designer avoids pairing metals far apart in the galvanic series when developing an outdoor structure exposed to rainwater. Justify this design decision based on corrosion behavior.
30.	A laboratory setup connects two dissimilar metals in an electrolyte, and one metal shows visible dissolution while the other remains unchanged. Infer the electrochemical roles of the two metals and relate them to galvanic principles.
31.	Two metallic components, one made of magnesium and the other of iron, are electrically connected in a moist soil environment, where conductivity varies with seasonal rainfall and mineral content. Assess how electrolyte variation impacts corrosion behavior and identify which metal is more vulnerable under these conditions.

32.	A ship hull constructed from steel includes small zinc plates attached intentionally and is continuously exposed to seawater, where these plates gradually dissolve over time while the main structure remains intact. Interpret the functional role of zinc plates and infer how their placement influences the corrosion protection mechanism.
33.	A pipeline system transporting saline water incorporates stainless steel valves connected to carbon steel pipes, where protective coatings are applied but periodically damaged due to mechanical wear. Field data indicates accelerated corrosion near the connection points, especially when flow stagnation occurs. Engineers suspect that both material differences and environmental factors are contributing to failure. Determine the governing factors leading to corrosion in this system and recommend how the galvanic series and environmental control can be used to mitigate the issue.
34.	An industrial heat exchanger uses copper tubes mechanically joined with aluminum fins in a humid environment containing slightly acidic condensate. Over time, maintenance reports reveal pitting on aluminum surfaces near the joints, while copper remains relatively unaffected. The system operates continuously, and moisture retention increases during shutdown periods, forming an electrolyte film between the metals. Evaluate the corrosion mechanism occurring in the system and infer how environmental conditions and material positioning contribute to the degradation of one component over the other.
35.	A coastal engineering firm is selecting materials for a pier structure exposed to seawater, where steel piles are connected to bronze fittings and partially coated with protective paint that has degraded unevenly over time. The seawater acts as a conductive electrolyte, and inspection reports indicate localized metal loss near junction regions where two metals meet. The engineers also observe that larger steel areas are connected to smaller bronze components, and the dissolved oxygen concentration varies across the structure due to tidal movement. Analyze the electrochemical interactions within the given system and determine which metal will preferentially corrode, while justifying how the galvanic series and area ratio influence the observed degradation pattern.
36.	A corrosion control system uses magnesium anodes in a high-resistivity soil environment, but protection is found to be weak. Determine the cause of reduced protection and propose an improvement strategy.
37.	A steel water tank shows localized corrosion despite having sacrificial anodes attached at only one side of the structure. Infer the reason for incomplete protection and recommend a corrective measure based on electrochemical principles.
38.	An industrial facility considers switching from sacrificial anode protection to impressed current protection for buried pipelines due to rising maintenance costs. Assess the trade-offs involved in transitioning between these two methods and determine the conditions under which such a switch becomes advantageous.

39.	A coastal bridge uses aluminum sacrificial anodes for corrosion protection, but engineers observe that protection efficiency decreases in colder seasons. Evaluate how environmental conditions influence the performance of aluminum anodes and infer the underlying factors affecting corrosion control efficiency.
40.	A large oil storage tank is fitted with an impressed current cathodic protection system that includes a DC power source, inert anodes, and monitoring units. During operation, inconsistent protection levels are observed across different sections of the tank. Diagnose the possible causes of uneven protection in the system and formulate corrective strategies to ensure uniform corrosion prevention across the entire structure.
41.	A ship hull operating continuously in seawater is protected using zinc blocks attached to its surface. Over time, the zinc blocks show significant material loss while the steel structure remains intact. Interpret the electrochemical mechanism that enables zinc to protect the steel hull and determine how material properties influence the effectiveness of this corrosion prevention strategy.
42.	An underground steel pipeline transporting crude oil across a coastal region is experiencing accelerated corrosion due to saline soil conditions. Engineers propose using a cathodic protection system but must decide between sacrificial anode and impressed current methods while considering installation cost, maintenance frequency, and effectiveness over long distances. Analyze the suitability of selecting either sacrificial anode or impressed current cathodic protection for this pipeline by evaluating operational constraints, and justify the most effective method based on corrosion control efficiency and system scalability.
	Unit 4
1.	A reaction in a laboratory shows reactants at 100 kJ/mol and products at 40 kJ/mol, with a peak at 160 kJ/mol. The energy profile is plotted against reaction coordinate. The system releases energy as the reaction progresses. Scientists analyze energy change and activation barrier. Determine the nature of the reaction and evaluate both the energy change and activation energy from the given data, and justify how these influence reaction feasibility and progression.
2.	A reaction has reactants at 200 kJ/mol and products at 80 kJ/mol, but the peak reaches 350 kJ/mol. The reaction proceeds very slowly. The energy diagram shows a steep rise before the peak. Scientists investigate the cause of slow rate. Analyze the discrepancy between favorable energy change and slow reaction rate, and infer the dominant factor controlling the reaction progression.
3.	An industrial reaction shows reactants at 150 kJ/mol, products at 90 kJ/mol, and a peak at 300 kJ/mol. The process is slow and energy intensive. Engineers aim to improve efficiency without altering products. They evaluate modification strategies. Recommend a suitable strategy to enhance reaction rate and justify how it alters the energy profile and reaction behavior.

4.	Two reactions are compared where one has a low peak but small energy difference, and another has a high peak but large energy difference. Both occur under identical conditions. Scientists evaluate speed and stability. Integrate the concepts of activation energy and energy difference to determine which reaction is faster and which is more stable.
5.	A factory compares two processes where one has low activation energy but low energy output, and another has high activation energy with high energy output. The aim is continuous production. Engineers assess efficiency. Examine the trade-offs and prioritize the more efficient process for large-scale production.
6.	A student draws an energy profile where reactants are higher in energy than products with a single peak. The graph represents energy versus reaction coordinate. The student must interpret reaction nature. Infer the reaction type and justify using the energy profile characteristics.
7.	A reaction shows a very high peak followed by a steep drop in energy. The process releases large energy quickly. Safety concerns arise in handling this reaction. Engineers evaluate risks. Assess the reaction suitability considering both safety and feasibility.
8.	A chemical plant studies a reaction where reactants must cross an energy barrier before forming products. Engineers observe that even though the reaction is thermodynamically favorable, it proceeds slowly at room temperature. They analyze the energy profile diagram and identify a peak representing a high-energy intermediate state. Additional heating significantly increases the reaction rate. Analyze the role of the energy barrier in controlling the reaction progression and determine how the identified intermediate state influences the reaction rate under varying temperature conditions. Justify the observed rate enhancement with temperature.
9.	In a laboratory, a reversible reaction is monitored where both forward and reverse reactions occur. The forward reaction requires less energy compared to the reverse process. Scientists plot an energy diagram and observe different activation energies for both directions, along with distinct energy levels for reactants and products. Determine how the difference in forward and reverse activation energies governs the directionality and equilibrium behavior of the reaction. Infer the implications for reaction feasibility and stability.
10.	A researcher compares two reactions: one involving strong covalent bond breaking and another involving weaker intermolecular interactions. The reaction with strong bonds proceeds significantly slower despite similar conditions. Molecular orientation during collisions is also noted to affect reaction success. Analyze how bond strength and molecular orientation collectively influence activation energy and determine their impact on the observed rate differences between the two reactions.

11.	During an experiment, a chemist observes that increasing temperature accelerates a reaction significantly. The reaction involves molecules overcoming an energy barrier to form products, and the distribution of molecular energies changes with temperature. Interpret how temperature variation alters molecular energy distribution and determine its effect on activation energy crossing probability.
12.	A reaction mechanism study identifies a short-lived, unstable configuration formed during the reaction pathway. This configuration exists momentarily at the highest energy point before converting into products. Infer the significance of this unstable configuration in the reaction pathway and determine its relationship with activation energy.
13.	A reaction profile diagram shows reactants at a certain energy level and products at a lower energy level, with a noticeable peak between them. Determine the significance of the peak in the diagram and infer its role in reaction kinetics
14.	Two reactions occur at the same temperature, but one proceeds much faster due to a lower energy requirement for bond rearrangement. Analyze the reason for the difference in reaction speeds based on energy considerations.
15.	A chemical plant is producing ammonia using a reaction that proceeds very slowly under standard conditions. Engineers introduce an iron-based catalyst into the system and observe that the reaction rate increases significantly without increasing temperature or pressure. The catalyst remains unchanged after multiple cycles. Energy profile analysis shows a reduced peak for the catalyzed pathway. Analyze how the introduction of the catalyst modifies the reaction pathway and justify the observed increase in reaction rate using energy considerations and catalyst properties.
16.	In a pharmaceutical reaction, a specific enzyme catalyzes only one type of substrate while ignoring structurally similar molecules. The reaction proceeds rapidly in small quantities of the enzyme, and repeated cycles show no loss in activity. The system reaches the same equilibrium position as before catalysis. Interpret the behavior of the enzyme in terms of specificity, efficiency, and its influence on equilibrium, and justify how these factors contribute to industrial applicability.
17.	A reaction system is studied with and without a catalyst, and energy profile diagrams are plotted. The catalyzed reaction shows a lower peak while both reactions start and end at the same energy levels. The rate of forward and reverse reactions increases equally in the presence of the catalyst. Compare the catalyzed and uncatalyzed reactions using the energy profile data and determine the effect of the catalyst on both reaction kinetics and equilibrium conditions.

18.	A laboratory reaction uses a catalyst that completes 500 reaction cycles per minute. The amount of catalyst used is very small compared to reactants, and the reaction proceeds efficiently without catalyst degradation. Researchers calculate a performance metric based on reaction cycles per unit time. Determine the significance of turnover number in this system and infer how it reflects catalytic performance.
19.	An industrial reaction initially requires high temperature to proceed at a reasonable rate. After introducing a catalyst, the same reaction occurs quickly at a lower temperature, reducing energy consumption. No change in product yield at equilibrium is observed. Analyze how the catalyst contributes to energy savings and infer why equilibrium yield remains unaffected.
20.	A reaction mixture contains a catalyst that accelerates only one specific reaction among several possible side reactions. As a result, undesired products are minimized. The catalyst is recovered after completion. Infer the role of catalyst selectivity in controlling product formation and justify its importance.
21.	A biological system uses enzymes to regulate metabolic reactions efficiently under mild conditions. These enzymes function repeatedly and maintain reaction rates necessary for survival. Determine how biological catalysts maintain efficiency under mild conditions and justify their functional advantage.
22.	A liquid-phase reaction is carried out where both the catalyst and reactants are dissolved in the same solvent. During the reaction, an intermediate complex forms temporarily before producing the final product. The rate of reaction depends on both catalyst concentration and reactant concentration. The process shows high selectivity but requires complex separation afterward. Analyze the reaction pathway and determine how the presence of an intermediate influences the kinetics and overall efficiency of the catalytic system, while also justifying the benefits and limitations observed.
23.	In an industrial reactor, an acid is used to catalyze an esterification reaction in a liquid medium. The catalyst interacts with the reactant to form a protonated intermediate. The rate increases significantly with increasing acid concentration but stabilizes after a limit. The system operates under controlled temperature and solvent conditions. Interpret the catalytic mechanism and formulate the relationship between catalyst concentration and reaction rate, while evaluating the role of solvent in controlling reaction behavior.

24.	A homogeneous catalytic reaction is monitored where decomposition of a compound occurs in solution with a dissolved catalyst. Experimental data shows that the rate law includes both reactant and catalyst concentration terms. The reaction proceeds via multiple elementary steps and produces consistent yields. Analyze the kinetic interpretation of the system and determine how the stepwise mechanism affects rate law formation and overall reaction control.
25.	A reaction occurs in a single liquid phase where catalyst and reactants are fully miscible. The system shows rapid reaction rates and formation of a temporary species before final products are obtained. Infer the role of phase uniformity and intermediate formation in determining the reaction efficiency and pathway.
26.	A homogeneous catalytic reaction exhibits dependency on solvent polarity and catalyst concentration. Changing the solvent alters the reaction rate and product distribution. Analyze how solvent properties and catalyst concentration collectively influence the reaction kinetics and selectivity.
27.	A solution-based catalytic reaction produces high yields but requires additional steps to recover the catalyst after completion. Determine the trade-off between efficiency and practicality in this catalytic system.
28.	A homogeneous reaction shows that increasing catalyst concentration beyond a point does not significantly increase the reaction rate. Interpret the kinetic limitation observed in the system.
29.	A chemical plant operates a reaction where gaseous reactants pass over a solid catalyst bed. The engineers observe that increasing the surface roughness of the catalyst improves the reaction rate significantly. However, beyond a certain point, further increase in temperature reduces efficiency due to unwanted side effects. The process involves adsorption of reactants, surface reaction, and product release. Analyze the sequence of steps occurring in this catalytic system and determine how surface properties and temperature collectively influence the overall reaction efficiency. Justify the dominant stage controlling the rate under the given conditions.
30.	An industrial ammonia synthesis unit uses a solid iron catalyst. Engineers detect that impurities in the feed gas reduce the catalytic activity over time. Despite maintaining optimal pressure and temperature, the conversion efficiency drops steadily. Surface analysis shows blockage of certain reactive regions. Analyze the cause of efficiency loss and determine how surface interactions influence catalyst performance. Recommend a reasoning-based strategy to restore catalytic activity.

31.	A research team studies two catalysts for the same reaction: one finely powdered and another in large solid chunks. Both are chemically identical but show different efficiencies under identical conditions. The powdered catalyst performs significantly better in terms of conversion rate. Interpret the observed difference in catalytic performance by integrating concepts of surface phenomena and reaction mechanisms. Formulate the relationship between catalyst structure and efficiency.
32.	In a catalytic converter, exhaust gases pass over a solid catalyst where harmful gases are converted into less toxic substances. At lower temperatures, the conversion is incomplete, while at optimal temperatures, efficiency peaks. Determine the role of temperature in influencing adsorption and reaction processes in this system and infer the reason for incomplete conversion at low temperatures.
33.	A laboratory experiment compares physisorption and chemisorption of gases on a solid surface. One process is found to be reversible and weak, while the other is strong and often irreversible. Analyze both adsorption types and determine their suitability in catalytic reactions based on interaction strength and reversibility.
34.	A catalytic reaction slows down despite sufficient reactant supply. Investigation shows accumulation of products on the catalyst surface. Infer the stage causing inefficiency and determine its impact on the catalytic cycle.
35.	A reaction system increases pressure to enhance catalytic efficiency in a gas-solid reaction. Determine how pressure influences the catalytic process and justify its effect on reaction rate.
36.	A chemical manufacturing unit observes that adding a small quantity of a substance increases the reaction rate significantly, while another similar-looking additive slows it down. The process engineers also note that the rate enhancement does not change the final equilibrium composition. They aim to classify these effects properly and optimize the process conditions. The system is sensitive to impurities, and small variations lead to large performance differences. Analyze the observed behavior and determine the classification of the substances affecting the reaction rate, and justify how their roles influence reaction pathways and equilibrium without altering final composition.

37.	In a laboratory reaction, scientists observe that one of the products formed during the reaction itself accelerates the reaction rate over time. Initially, the reaction is slow, but it becomes progressively faster without external intervention. The reaction medium remains unchanged, and no external catalyst is added. Researchers aim to model this unusual behavior. Interpret the reaction behavior and formulate the catalytic classification involved, and infer how the evolving system dynamics influence rate progression and reaction mechanism.
38.	An industrial plant compares two catalytic systems: one where reactants and catalyst are in the same phase, and another where the catalyst is solid while reactants are gases. The first system shows uniform reaction but is difficult to separate, while the second system shows surface-based reactions with easier separation. The engineers must select the best system for large-scale production. Examine both catalytic systems and determine their comparative advantages and limitations in terms of mechanism, efficiency, and industrial feasibility, leading to a justified selection.
39.	A reaction system shows improved performance when a small additive is introduced along with the catalyst. However, when another impurity enters the system, the catalytic activity drops significantly. Engineers suspect these substances are influencing catalyst efficiency in opposite ways. Determine the roles of the added substances and infer their impact on catalytic efficiency and reaction performance under the given conditions.
40.	A biochemical reaction proceeds rapidly only in the presence of a specific protein, and even slight structural changes in the protein reduce the reaction rate drastically. The reaction occurs under mild conditions and shows high specificity toward a single substrate. Analyze the catalytic system and infer the nature of catalysis and its dependence on structural and environmental factors.
41.	A chemical process is modified to reduce harmful emissions by replacing traditional catalysts with nanoscale materials. The new system shows higher efficiency and reduced waste generation under similar conditions. Infer the type of catalytic advancement involved and determine its contribution to environmental performance.
42.	A reaction in an energy system requires selective conversion of one reactant among many similar compounds. The catalyst used ensures only the desired product forms while minimizing side reactions. Determine the catalytic property responsible and infer its significance in industrial applications.
	Unit 5

1.	A nuclear model shows that increasing neutron number beyond a limit leads to instability in certain elements. Interpret why excessive neutrons reduce nuclear stability.
2.	Two atoms have the same atomic number but different atomic masses, yet they react identically in chemical reactions. Justify why their chemical behavior remains unchanged despite mass differences.
3.	A physicist calculates the mass of a nucleus and compares it with the sum of individual nucleon masses, finding a discrepancy. This difference is associated with energy release during nucleus formation. Determine the significance of this mass discrepancy and relate it to nuclear binding energy.
4.	A laboratory detects two isotopes of the same element where one is used in medical diagnostics due to its emission properties, while the other is non-reactive. Both have identical electron configurations but differ in nuclear composition. Infer how nuclear differences enable one isotope to be suitable for diagnostic applications while the other remains inactive.
5.	A nuclear engineer studies a set of isotopes plotted on a neutron-to-proton ratio graph and notices that certain nuclei lie within a stability band while others fall outside it. The unstable nuclei tend to undergo decay processes to reach equilibrium. Observations indicate a pattern relating neutron excess to instability. Interpret the relationship between neutron-to-proton ratio and nuclear stability and determine why nuclei outside the stability band tend to transform.
6.	An energy researcher compares two uranium samples used in nuclear applications, observing that one sustains controlled reactions while the other remains relatively inactive. Both samples share identical atomic numbers but exhibit differences in mass and energy output potential. Experimental observations suggest variations in nuclear binding forces. Evaluate how differences in nuclear composition influence binding energy and justify the observed variation in energy release capability between the two samples.
7.	A materials scientist is analyzing two carbon samples extracted from different environments, where both samples show identical chemical behavior but differ slightly in mass measurements. One sample remains unchanged over time while the other undergoes gradual transformation. Instrumental data reveals variation in neutron count despite identical proton numbers. The scientist aims to understand the structural cause behind this difference and its implications. Analyze the structural variation between the two samples and determine how nuclear composition influences their stability and transformation behavior under identical chemical conditions.
8.	In a diffusion experiment, scientists compare rates of gas flow for isotopes under identical pressure conditions. The lighter isotope consistently moves faster. Analyze the relationship between mass and diffusion rate from the observed data.
9.	Two isotopes of the same element are heated and their boiling points are measured. A slight variation is observed despite identical chemical composition. Determine the reason for variation in boiling point and relate it to atomic characteristics.

10.	A research team uses isotopes as tracers to track water movement in soil. The isotopes behave chemically identical to normal water molecules but can be detected separately using instruments. This allows monitoring of flow patterns without altering the system. Analyze how isotopic properties enable tracing applications and infer why they do not disturb the system behavior.
11.	An industrial setup uses centrifugation to separate uranium isotopes. The heavier isotope moves outward more strongly under rotational force, while the lighter one remains closer to the center. The system operates at high rotational speeds to enhance separation efficiency. Determine how centrifugal force enables isotope separation and determine the role of mass difference in the process.
12.	A pharmaceutical company studies reaction rates using isotopically labeled compounds and notices that replacing hydrogen with deuterium slows down certain reactions. The bond strength appears slightly altered, affecting reaction kinetics. The goal is to understand how isotopic substitution influences reaction mechanisms in drug synthesis. Interpret the influence of isotopic substitution on reaction rate and justify the observed behavior using underlying kinetic principles.
13.	In a laboratory experiment, two isotopes of hydrogen are allowed to diffuse through a porous membrane. The lighter isotope escapes faster than the heavier one under identical experimental conditions. Researchers aim to use this observation to design an isotope separation system based on diffusion principles. Temperature and pressure are maintained constant throughout the experiment. Determine the governing principle behind the observed diffusion difference and evaluate how it can be applied to isotope separation in industrial systems.
14.	A chemical plant compares two isotopes of chlorine used in parallel reactions and observes identical reaction pathways but slight differences in reaction rates. The engineers suspect that while electron configurations remain unchanged, atomic mass may influence certain measurable parameters. They aim to evaluate how isotopic substitution impacts both chemical and physical behavior in controlled conditions. The system operates under constant temperature and pressure with identical reactant concentrations. Analyze the observed similarities and differences in the behavior of the isotopes and determine how atomic structure and mass contribute to these variations under the given controlled conditions.
15.	A decay series shows successive emissions where mass number decreases significantly at one stage and remains constant at another stage. Researchers track these variations to identify decay types. Interpret the decay stages and correlate them with corresponding radiation types.
16.	A radioactive nucleus emits radiation that does not alter its atomic structure but releases energy. The radiation passes through thick materials easily and shows no charge. Determine the emission type and infer its role in nuclear stability.

17.	An artificially produced radioactive isotope is observed to decay by emitting radiation that increases the atomic number of the daughter nucleus. The radiation shows noticeable deflection in electric fields and moderate penetration ability. Analyze the decay mechanism and determine how artificial radioactivity leads to such nuclear transformation.
18.	A radioactive substance is placed near a detector that measures ionization levels. The radiation produces very strong ionization but fails to pass through a thin sheet of paper. The emitted particles are suspected to be massive and charged. Infer the type of radiation and justify the relationship between its ionizing power and penetration capability.
19.	A medical imaging facility uses a radioactive source that emits highly penetrating radiation with no charge and no mass. The radiation is not deflected in electric or magnetic fields. The nucleus emitting this radiation does not change its atomic or mass number. Researchers aim to interpret this emission type. Identify the radiation type and evaluate its nuclear significance. Interpret why no change occurs in nuclear composition during this emission.
20.	In a nuclear research setup, a radioactive isotope emits fast-moving electrons that show strong deflection in a magnetic field. The emitted radiation has moderate penetration and moderate ionization effects. The nucleus formed after emission shows an increase in atomic number but unchanged mass number. Scientists are evaluating the decay mechanism. Analyze the decay process involved and infer how nuclear composition changes. Reconstruct the reasoning linking particle properties to nuclear transformation.
21.	A laboratory is analyzing an unknown radioactive sample that emits heavy particles with low penetration but high ionizing power. The emitted particles are observed to be deflected slightly in an electric field. The nucleus of the atom changes significantly after emission. Researchers are trying to identify the decay type and its nuclear transformation. Determine the nature of the emitted radiation and analyze how the atomic number and mass number of the parent nucleus transform during this decay process. Justify your conclusion using the observed properties.
22.	A scientist studies the average lifetime of radioactive nuclei in a sample. The decay constant is known and used to interpret the duration of existence of nuclei. The scientist wants to relate average lifetime with half-life. This helps in understanding decay duration more clearly. The analysis is conceptual. Analyze the relationship between mean life and half-life and determine how they are connected through decay behavior.

23.	A radioactive sample reduces to one-fourth of its original activity over a period of time. The researcher observes that the reduction follows a consistent halving pattern. This helps in tracking decay progression. The focus is on interpreting the number of decay stages. The observation is based on fractional decrease. Determine how many decay stages are involved when activity reduces to one-fourth of its initial value.
24.	A medical physicist observes radioactive tracers under different environmental conditions such as temperature and pressure. The decay rate remains unchanged in all cases. Repeated experiments confirm identical results. The physicist evaluates whether external factors influence decay. The goal is to ensure reliability in applications. Analyze the influence of environmental conditions on radioactive decay and justify the observed consistency.
25.	A graph plotted between number of nuclei and time shows a continuously decreasing curve. The decrease is rapid initially and becomes slower later. The curve never reaches zero even after long durations. Scientists study this graphical trend to understand decay behavior. The aim is to interpret its meaning Interpret the graphical representation of radioactive decay and determine what it indicates about the variation in decay rate.
26.	An archaeologist studies an ancient object containing radioactive material whose activity has significantly decreased over time. The decay follows a predictable pattern and provides a way to estimate the age of the object. The observed activity is compared with expected initial levels. The analysis depends on understanding decay behavior over long durations. The goal is to infer age from decay trends. Analyze how the concept of half-life can be applied to infer the age of the artifact based on observed reduction in activity.
27.	A radioactive material in a laboratory consistently reduces to half of its initial amount over equal intervals of time. Researchers compare different isotopes based on this halving behavior. They observe that some isotopes decay faster while others persist longer. The comparison depends on internal decay characteristics. The aim is to relate halving time with decay nature. Interpret the relationship between half-life and decay constant and determine how this relationship enables comparison of different radioactive materials.

28.	A nuclear laboratory monitors a radioactive isotope used in imaging, where the activity decreases steadily over time. Scientists observe that the rate of disintegration depends on the number of undecayed nuclei present at any instant. The decay pattern shows a smooth and continuous reduction. The team wants to understand the governing principle behind this behavior. They aim to interpret how decay constant controls the process. Analyze the governing principle of radioactive decay and determine how the decay constant influences the rate of disintegration in the given system.
29.	Two nuclear processes are compared: one splits heavy nuclei, while the other combines light nuclei. Both release energy but differ in operational conditions. Differentiate the processes based on mechanism and operational requirements.
30.	A system requires a minimum amount of fissile material to sustain a chain reaction. Below this limit, reactions stop quickly. Above it, reactions grow rapidly. Determine the governing concept and infer its role in reaction sustainability.
31.	A nuclear reaction releases energy when the total mass of products is lower than the reactants. Scientists compare this with another reaction where energy must be supplied to proceed. Both reactions are studied for efficiency. Analyze the role of mass-energy relationships in determining whether a reaction releases or absorbs energy.
32.	A reactor core contains fuel rods surrounded by a material that slows down neutrons. Another set of rods can be inserted or removed to regulate the reaction. The system must operate at steady power without overheating. Engineers monitor neutron flux continuously. Infer how different reactor components collaboratively maintain controlled nuclear reactions and prevent instability.
33.	A nuclear experiment tracks a reaction where a target nucleus is bombarded by particles, resulting in a different element being formed. The reaction obeys strict conservation of mass number and atomic number. Scientists calculate the energy released using measured mass differences. The reaction is being optimized for isotope production. Formulate the governing principles that validate the reaction representation and infer how energy changes are evaluated using measurable quantities.

34.	In a high-temperature experimental setup, two light nuclei are forced to combine under extreme pressure, producing a heavier nucleus and releasing substantial energy. The system requires continuous input of energy to overcome repulsive forces between nuclei. Scientists are evaluating whether this process can be scaled for long-term energy production. Measurements indicate significantly higher energy output compared to conventional methods. Determine the nature of the reaction and justify the energy behavior by integrating nuclear forces, energy transformations, and feasibility considerations for sustained application.
35.	A research facility is evaluating a heavy nucleus that splits into two smaller fragments after absorbing a neutron, releasing additional neutrons and energy. The system is designed to sustain the reaction under controlled conditions using specific materials to regulate neutron speed. The engineers are concerned about maintaining stability while ensuring continuous energy output. Observations show that slight increases in neutron population rapidly amplify the reaction. Analyze the reaction process and determine the governing mechanism that sustains it, integrating how control strategies regulate the reaction rate and ensure stability in the system.
36.	Environmental scientists measure isotope concentrations in water bodies to study pollution sources. Different pollutants carry distinct isotopic signatures depending on origin. Infer how isotopic analysis helps identify pollution sources in environmental systems.
37.	An industrial plant uses isotopes to detect leaks in pipelines by introducing traceable radioactive materials and monitoring their escape points. Sensors detect radiation levels externally. Determine how isotope tracing enables leak detection and justify its efficiency.
38.	Geologists study rock formations using isotopes with long half-lives. They compare parent and daughter isotope ratios to estimate the age of Earth materials. The system is assumed to remain closed over geological time. Analyze how isotope ratios contribute to geological dating and infer the importance of system closure.

39.	Medical researchers use radioactive isotopes to trace metabolic pathways in the human body. The isotopes emit detectable radiation without significantly altering biological processes. Monitoring devices track their movement through organs. Assess how isotopes function as tracers and justify their effectiveness in medical diagnostics.
40.	A laboratory analyzes fossil samples but finds discrepancies in calculated ages due to environmental contamination and variations in historical carbon-14 levels. Scientists introduce calibration curves and correction factors based on tree-ring data and known-age samples. Examine the limitations of radiocarbon dating and recommend how calibration techniques improve accuracy in such cases.
41.	Atmospheric scientists observe that cosmic rays continuously interact with nitrogen atoms, producing carbon-14. This isotope mixes with atmospheric carbon dioxide and enters plant systems through photosynthesis. Animals then consume these plants, maintaining a dynamic equilibrium of carbon isotopes while alive. Interpret the formation and distribution mechanism of carbon-14 and infer how this continuous cycle supports its use in chronological studies.
42.	A team of archaeologists recovers wooden artifacts from an ancient settlement buried under sediment layers. They measure the remaining carbon-14 content and compare it with modern atmospheric levels. The decay constant and half-life of carbon-14 are known. The team assumes the organism stopped exchanging carbon after death and that decay followed a predictable exponential pattern. Analyze the process by which the age of the artifact can be determined and justify how the decay principle enables chronological estimation under the given assumptions.